

# Pituitary and Hormonal Axis

Growth hormone and IGF-1 (GH1, GHR, IGF1, IGFBP3), prolactin and dopamine (PRL, DRD2), the gonadal axis (FSHB, FSHR) and pituitary adenomas (AIP)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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# 1. Summary

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**Favourable. Genetic IGF-1 is low to typical:** the consensus of the 5 scores with good coverage is at percentile **32.1**, the genomic scores at percentile 28.8, and the burden of 807 IGF-1 loci at **21.6**. The main **IGF1** locus lacks the raising allele (A/A), and so does the IGF1 promoter (G/G). In longevity data, lower IGF-1 is associated with longer life, and **FOXO3 rs2802292 is G/G** (2 longevity alleles in the same pathway). Male testosterone has typical-to-high genetics: the consensus is at percentile **58.0** and the most recent genomic score at percentile 89.3. For pituitary adenomas, **AIP R304Q** is absent (G/G).

**Unfavourable / to watch.** Low IGF-1 has the other side of the balance: less anabolic stimulus for muscle and bone, in line with the genetic lean mass at the 27th percentile in other reports. Prolactin has 3 loci with a burden at percentile 88.4 (JAK2 C/C), but the Hormones report, with another set of loci, gave the 12th percentile; the reading is inconclusive. **FSHB -211T** is present in 1 copy (G/T), which slightly lowers FSH.

**What it means.** The pituitary-IGF-1 axis is on the slower side of the distribution, the profile that longevity studies associate with longer life, with genetically typical-to-high testosterone helping the muscle. Actual IGF-1 depends heavily on dietary protein, sleep, training and insulin; the value measured in blood tests is what counts. There are no genetic signs of pituitary disease (hereditary adenomas depend on rare AIP and MEN1 mutations, which the chip does not read beyond R304Q).

## 2. How to read this report

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- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality DR2  $\geq 0.8$ . Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 15 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 13 from the chip, 2 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 4 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 5,619,814 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ( $n = 633$ ), calculated with exactly the same variants. Percentile  $\geq 80$ : MORE LIKELY/HIGH;  $\leq 20$ : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with  $p \leq 5 \times 10^{-8}$  in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ( $n = 633$ ) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 883 loci, 768 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

### 3. Polygenic scores

**Consensus by trait.** For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage  $\geq 75\%$** , all calibrated in the same reference population ( $n = 633$ ). The genomic column is the median of the scores with  $\geq 100,000$  variants. Scores with coverage  $< 75\%$  are shown in the tables but do not enter the consensus.

| Group              | Consensus (percentile) | Scores with coverage $\geq 75\%$ | Genomic scores (percentile) | Genomic scores |
|--------------------|------------------------|----------------------------------|-----------------------------|----------------|
| IGF-1              | 32.1                   | 5 of 5                           | 28.8                        | 3              |
| Growth hormone     | —                      | 0 of 1                           | —                           | 0              |
| Testosterone (men) | 58.0                   | 3 of 3                           | 69.2                        | 2              |
| SHBG               | 87.2                   | 1 of 1                           | 87.2                        | 1              |

For IGF-1 and growth hormone the percentile is descriptive (no judgement): lower IGF-1 favours longevity but gives less stimulus to muscle. The growth hormone score has 33% coverage and is left out of the consensus. For testosterone, a higher percentile is favourable; SHBG is descriptive.

#### IGF-1

| Trait           | Score (PGS Catalog)                           | Variants used                         | Percentile | Result  | Judgement    |
|-----------------|-----------------------------------------------|---------------------------------------|------------|---------|--------------|
| IGF-1 (genomic) | PGS004338 (Jung 2024, 1.06 million variants)  | 1,025,639 of 1,059,939 (coverage 96%) | 75.7       | TYPICAL | no judgement |
| IGF-1 (genomic) | PGS003539 (Ding 2023, 980,000 variants)       | 972,627 of 979,739 (coverage 99%)     | 26.4       | TYPICAL | no judgement |
| IGF-1 (genomic) | PGS002178 (Privé 2022, 818,000 variants)      | 802,757 of 817,740 (coverage 98%)     | 28.8       | TYPICAL | no judgement |
| IGF-1           | PGS001960 (Privé 2022, 87,000 variants)       | 83,731 of 87,149 (coverage 96%)       | 32.1       | TYPICAL | no judgement |
| IGF-1           | PGS000687 (PGS Catalog 2021, 23,000 variants) | 20,829 of 23,442 (coverage 90%)       | 34.1       | TYPICAL | no judgement |

#### Growth hormone

| Trait                          | Score (PGS Catalog)                          | Variants used                     | Percentile | Result | Judgement    |
|--------------------------------|----------------------------------------------|-----------------------------------|------------|--------|--------------|
| Blood growth hormone (genomic) | PGS000244 (Folkersen 2020, 552,000 variants) | 190,424 of 552,236 (coverage 33%) | 3.8        | LOW    | no judgement |

#### Testosterone (men)

| Trait                       | Score (PGS Catalog)                            | Variants used                     | Percentile | Result  | Judgement  |
|-----------------------------|------------------------------------------------|-----------------------------------|------------|---------|------------|
| Testosterone (men, genomic) | PGS003559 (PGS Catalog 2023, 980,000 variants) | 972,627 of 979,739 (coverage 99%) | 89.3       | HIGH    | favourable |
| Testosterone (men, genomic) | PGS002205 (Privé 2022, 585,000 variants)       | 574,696 of 584,991 (coverage 97%) | 49.0       | TYPICAL | typical    |
| Testosterone (men)          | PGS001988 (Privé 2022, 3,985 variants)         | 3,857 of 3,985 (coverage 97%)     | 58.0       | TYPICAL | typical    |

#### SHBG

| Trait          | Score (PGS Catalog)                            | Variants used                     | Percentile | Result | Judgement    |
|----------------|------------------------------------------------|-----------------------------------|------------|--------|--------------|
| SHBG (genomic) | PGS003552 (PGS Catalog 2023, 980,000 variants) | 972,627 of 979,739 (coverage 99%) | 87.2       | HIGH   | no judgement |

## 4. Key variants by topic

Variants of growth hormone and IGF-1 (IGF1, IGFBP3, GHSR, GHR, FOXO3), of prolactin and its dopaminergic brake (PRL, DRD2), of the gonadal axis (FSHB, FSHR, LHCGR) and of pituitary adenoma genes (AIP, CDKN1B). Genes with only a quantitative effect receive no judgement.

### Growth hormone and IGF-1

IGF-1 is made in the liver under the control of growth hormone and is the main anabolic signal; the insulin/IGF-1 pathway is one of the most studied longevity targets.

| Gene   | SNP        | Variant | Genotype | Source              | Effect allele (copies) | Judgement    | Allele effect                                                                                                                                                                                             | Evidence |
|--------|------------|---------|----------|---------------------|------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| IGF1   | rs35767    |         | G/G      | chip                | T (0 copies)           | no judgement | IGF1 promoter: the T allele is associated with higher circulating IGF-1                                                                                                                                   | Moderate |
| IGF1   | rs11111274 |         | A/A      | imputed<br>DR2=0.99 | G (0 copies)           | no judgement | IGF1 region: main locus of IGF-1 in blood [GWAS Catalog: IGF 1 (UKB data field 30770), allele A, beta 0.068 [0.064-0.072] unit decrease; risk/increase allele = G, the opposite, p=5E-239, PMID 39789286] | Strong   |
| IGFBP3 | rs2854744  | A-202C  | G/T      | chip                | C (1 copy)             | no judgement | A-202C of IGFBP3: the C allele lowers IGFBP-3 (protein transporting IGF-1)                                                                                                                                | Moderate |
| IGFBP3 | rs11977526 |         | G/A      | chip                | A (1 copy)             | no judgement | IGFBP3 region: IGF-1 and IGFBP-3 [GWAS Catalog: IGF-1 and IGFBP-3 levels (bivariate analysis), allele A, p=3E-195, PMID 27329260]                                                                         | Moderate |
| GHSR   | rs572169   |         | C/T      | chip                | T (1 copy)             | no judgement | Ghrelin receptor (growth hormone secretagogue receptor) [GWAS Catalog: IGF 1 (UKB data field 30770), allele T, beta 0.047 [0.043-0.051] unit increase, p=2E-131, PMID 39789286]                           | Moderate |
| GHR    | rs6180     | I526L   | A/A      | chip                | C (0 copies)           | no judgement | I526L of the growth hormone receptor: the C allele is associated with slightly different IGF-1; small effect                                                                                              | Weak     |
| FOXO3  | rs2802292  |         | G/G      | chip                | G (2 copies)           | favourable   | FOXO3 (insulin/IGF-1 pathway): the G allele is associated with exceptional longevity in several cohorts                                                                                                   | Moderate |

### Prolactin and dopamine

Dopamine from the hypothalamus is the natural brake on prolactin.

| Gene           | SNP       | Variant | Genotype | Source | Effect allele (copies) | Judgement    | Allele effect                                                                                                                                             | Evidence |
|----------------|-----------|---------|----------|--------|------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| DRD2/<br>ANKK1 | rs1800497 | Taq1A   | G/G      | chip   | A (0 copies)           | no judgement | Taq1A (A1): the A allele is associated with fewer dopamine D2 receptors; dopamine is the brake of prolactin (more elevated prolactin with antipsychotics) | Moderate |
| PRL            | rs1341239 |         | A/C      | chip   | T (1 copy)             | no judgement | Extra-hypophyseal promoter of prolactin (-1149G/T): the T allele is associated with more prolactin in peripheral tissues                                  | Weak     |

| Gene | SNP       | Variant | Genotype | Source              | Effect allele (copies) | Judgement    | Allele effect                                                                                                                                                                                                  | Evidence |
|------|-----------|---------|----------|---------------------|------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| JAK2 | rs7019858 |         | C/C      | imputed<br>DR2=1.00 | C (2 copies)           | no judgement | JAK2 region (prolactin receptor signal): prolactin levels [GWAS Catalog: PRL protein levels, allele T, beta 0.069 [0.057-0.081] unit decrease; risk/increase allele = C, the opposite, p=2E-32, PMID 39789286] | Weak     |

## Gonadal axis (FSH and LH)

Pituitary FSH and LH drive the testis (links to the Testes, Hormones and Male Fertility reports).

| Gene  | SNP        | Variant | Genotype | Source | Effect allele (copies) | Judgement    | Allele effect                                                                                      | Evidence |
|-------|------------|---------|----------|--------|------------------------|--------------|----------------------------------------------------------------------------------------------------|----------|
| FSHB  | rs10835638 |         | G/T      | chip   | T (1 copy)             | no judgement | −211G>T of FSHB: the T allele lowers FSH (less stimulation of Sertoli cells)                       | Moderate |
| FSHR  | rs6166     | N680S   | T/T      | chip   | C (0 copies)           | no judgement | N680S of the FSH receptor: the C allele (S680) is associated with a slightly lower response to FSH | Weak     |
| LHCGR | rs2293275  | N312S   | C/C      | chip   | G (2 copies)           | no judgement | N312S of the LH receptor: small effect on the testis response to LH                                | Weak     |

## Pituitary adenomas and cell cycle control

Genes of rare hereditary adenomas; the rare mutations are not read by the chip.

| Gene   | SNP         | Variant | Genotype | Source | Effect allele (copies) | Judgement    | Allele effect                                                      | Evidence |
|--------|-------------|---------|----------|--------|------------------------|--------------|--------------------------------------------------------------------|----------|
| CDKN1B | rs2066827   | V109G   | T/G      | chip   | G (1 copy)             | no judgement | V109G of p27Kip1 (CDKN1B, MEN4 gene): common variant, small effect | Weak     |
| AIP    | rs104894190 | R304Q   | G/G      | chip   | A (0 copies)           | favourable   | R304Q of AIP: rare variant of low penetrance in pituitary adenomas | Moderate |

## 5. GWAS loci burden

Sum of GWAS Catalog loci ( $p \leq 5 \times 10^{-8}$ ) for IGF-1, growth hormone and its receptor, prolactin and FSH/LH. Alleles are oriented towards higher hormone levels (no judgement) and the sum is compared with the reference population.

| Trait (GWAS Catalog)            | Loci (with genotype) | Sérgio F's effect alleles | Calibrated variants | Percentile | Judgement    |
|---------------------------------|----------------------|---------------------------|---------------------|------------|--------------|
| IGF-1 (levels)                  | 807 (713)            | 686 of 1426               | 712                 | 21.6       | no judgement |
| Growth hormone and its receptor | 42 (24)              | 28 of 48                  | 24                  | 73.1       | no judgement |
| Prolactin                       | 3 (3)                | 4 of 6                    | 3                   | 88.4       | no judgement |
| FSH and LH                      | 31 (28)              | 34 of 56                  | 28                  | 44.2       | no judgement |

## 6. Favourable

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- **Low-to-typical genetic IGF-1:** consensus at percentile **32.1**, loci burden at **21.6**; IGF1 without the raising alleles (A/A, G/G). Favourable for the insulin/IGF-1 longevity pathway. **Moderate.**
- **FOXO3 rs2802292 G/G:** 2 alleles associated with exceptional longevity. **Moderate.**
- **Male testosterone:** consensus at percentile **58.0**; the most recent genomic score at percentile 89.3. **Weak to moderate.**
- **AIP R304Q absent** (G/G); DRD2 Taq1A without the A1 allele (G/G), with normal D2 receptor density (the prolactin brake). Moderate (DRD2) and strong (AIP, for this variant).

## 7. Unfavourable

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- **Low IGF-1 and muscle:** less anabolic signalling, in line with the genetic lean mass at the 27th percentile. **Moderate.**
- **Prolactin:** burden of 3 loci at percentile 88.4 (JAK2 C/C, PRL A/C); disagrees with the Hormones report (12th percentile). **Weak.**
- **FSHB –211G>T** G/T: 1 copy of the FSH-lowering allele; **LHCGR N312S** C/C. Small effects on testicular stimulation. **Weak.**
- **Genetically high SHBG** (percentile 87.2): less free testosterone per unit of total testosterone; partly offset by total testosterone. **Weak.**

## 8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

| Mechanism                           | Genetic finding                                                                                                | Meaning             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------|
| Deregulated Nutrient Sensing        | Insulin/IGF-1 pathway: genetic IGF-1 at percentile 32.1 (consensus) and FOXO3 G/G                              | Favourable          |
| Endocrine Dysregulation             | Pituitary axis: low-to-typical IGF-1, typical-to-high testosterone, inconclusive prolactin, slightly lower FSH | Typical             |
| Lean Mass Loss                      | Less anabolic IGF-1 signalling; partly offset by testosterone and strength training                            | Unfavourable (mild) |
| Sarcopenia                          | IGF-1 and testosterone are the two main anabolic signals for muscle; mixed genetic balance                     | Conditional         |
| Altered Intercellular Communication | Hormonal signals from the pituitary to the liver, testis and adrenal                                           | Typical             |
| Osteoporosis                        | IGF-1 contributes to bone mass; lower genetic IGF-1 is a small contribution                                    | Conditional         |

## 9. Practical suggestions

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All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **IGF-1 and IGFBP-3 in blood tests** once, to see where the actual value lies; the longevity target is the lower-middle part of the reference range, without sacrificing muscle. *Suggestion (not applied).*
2. **Do not use growth hormone secretagogues or IGF-1** without a medical indication: they raise IGF-1 and work against the genetic longevity advantage. *Suggestion (not applied).*
3. **Protect muscle** with strength training and enough protein spread through the day; this delivers the anabolic signal where it is needed without chronically raising systemic IGF-1. *Suggestion (not applied).*
4. **Prolactin, LH, FSH and testosterone** only if there are symptoms (low libido, fatigue, changes in peripheral vision or persistent headaches) or medicines that act on dopamine. *Suggestion (not applied).*
5. **Already in the plan and protective:** good-quality sleep (most growth hormone is released in deep sleep), daily exercise and glucose control. *Suggestion (not applied).*

## 10. Limitations

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- The IGF-1 scores describe the blood level, not the action in tissues; the growth hormone score has low coverage.
- Prolactin has few known loci, and the two sets of loci (this report and the Hormones report) disagree.
- Pituitary adenomas depend mainly on rare or somatic mutations (AIP, MEN1, GNAS, USP8) that the chip does not read; only AIP R304Q was read.
- **LHCGR N312S**: the C/C genotype on the + strand corresponds to S312/S312 on the gene strand; the Male Fertility report read this variant as absent. The difference comes from the strand, and the effect is small.
- Actual IGF-1 depends heavily on diet, sleep, training and insulin.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

## 11. Methods and sources

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- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with  $DR2 \geq 0.8$ . No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; where the literature uses the gene strand (FSHR, LHCGR, IGFBP3), the letters appear complemented. Published alleles with a lowering effect were inverted to the raising direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (8 scores: IGF-1, growth hormone, testosterone and SHBG). Score =  $\sum \text{coefficient} \times \text{dosage}$ . The percentile is empirical in the reference population ( $n = 633$ ), using the same variants. Consensus = median of the scores with effect-size coverage  $\geq 75\%$ .
- **Loci burden:** GWAS Catalog ( $p \leq 5 \times 10^{-8}$ ), 4 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Sinnott-Armstrong 2021 (IGF-1), Willcox 2008 (FOXO3), Daly 2006 and Vierimaa 2006 (AIP), Grigороva 2008 (FSHB -211G>T), Thompson 1997 (DRD2 Taq1A), PGS Catalog, GWAS Catalog and ClinVar.

## 12. Annex: all variants analysed

| Topic                                     | Gene           | SNP         | Genotype | Source              | Effect allele | Copies | Strand                                                             | Interpretation                    |
|-------------------------------------------|----------------|-------------|----------|---------------------|---------------|--------|--------------------------------------------------------------------|-----------------------------------|
| Growth hormone and IGF-1                  | IGF1           | rs35767     | G/G      | chip                | T             | 0      | complement (– strand)                                              | 0 copies of the effect allele     |
| Growth hormone and IGF-1                  | IGF1           | rs11111274  | A/A      | imputed<br>DR2=0.99 | G             | 0      | + strand                                                           | 0 copies of the effect allele     |
| Growth hormone and IGF-1                  | IGFBP3         | rs2854744   | G/T      | chip                | C             | 1      | complement (– strand)                                              | 1 copy of the effect allele       |
| Growth hormone and IGF-1                  | IGFBP3         | rs11977526  | G/A      | chip                | A             | 1      | + strand                                                           | 1 copy of the effect allele       |
| Growth hormone and IGF-1                  | GHSR           | rs572169    | C/T      | chip                | T             | 1      | + strand                                                           | 1 copy of the effect allele       |
| Growth hormone and IGF-1                  | GHR            | rs6180      | A/A      | chip                | C             | 0      | + strand                                                           | 0 copies of the effect allele     |
| Growth hormone and IGF-1                  | FOXO3          | rs2802292   | G/G      | chip                | G             | 2      | + strand                                                           | 2 copies of the protective allele |
| Prolactin and dopamine                    | DRD2/<br>ANKK1 | rs1800497   | G/G      | chip                | A             | 0      | + strand                                                           | 0 copies of the effect allele     |
| Prolactin and dopamine                    | PRL            | rs1341239   | A/C      | chip                | T             | 1      | complement (– strand)                                              | 1 copy of the effect allele       |
| Prolactin and dopamine                    | JAK2           | rs7019858   | C/C      | imputed<br>DR2=1.00 | C             | 2      | + strand                                                           | 2 copies of the effect allele     |
| Gonadal axis (FSH and LH)                 | FSHB           | rs10835638  | G/T      | chip                | T             | 1      | + strand                                                           | 1 copy of the effect allele       |
| Gonadal axis (FSH and LH)                 | FSHR           | rs6166      | T/T      | chip                | C             | 0      | + strand                                                           | 0 copies of the effect allele     |
| Gonadal axis (FSH and LH)                 | LHCGR          | rs2293275   | C/C      | chip                | G             | 2      | complement (– strand)                                              | 2 copies of the effect allele     |
| Pituitary adenomas and cell cycle control | CDKN1B         | rs2066827   | T/G      | chip                | G             | 1      | + strand                                                           | 1 copy of the effect allele       |
| Pituitary adenomas and cell cycle control | AIP            | rs104894190 | G/G      | chip                | A             | 0      | homozygous for the reference allele (monomorphic site on the chip) | without the risk allele           |